

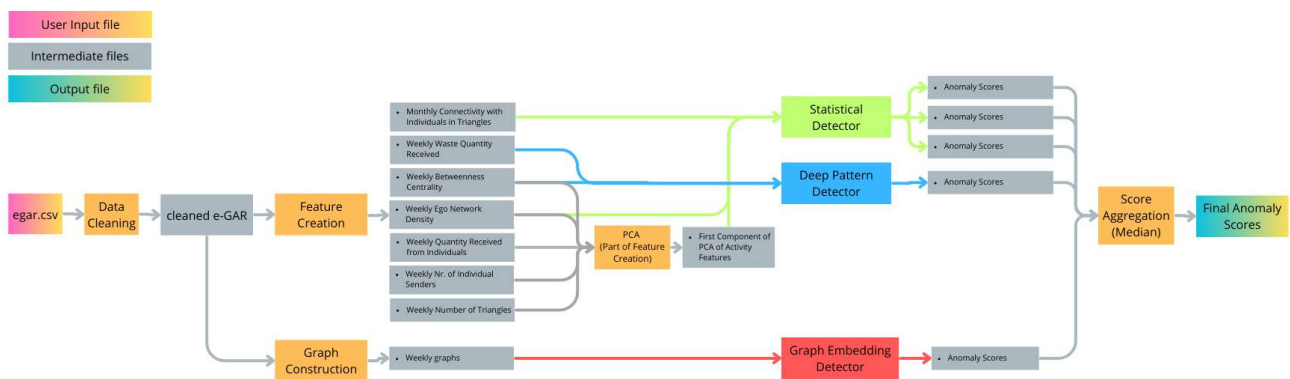
Overview of the Anomaly Detection Approach

This approach was designed to help identify companies that may be involved in illicit waste management activities. It analyzes the waste transport records (e-GAR) to detect unusual behavior that differs from what is normally observed in the waste management network.

Illegal practices in waste management often involve coordination between multiple actors, such as companies and individuals, and tend to occur sporadically rather than continuously, as a way to avoid detection. Because of this, simply looking at total volumes or isolated transactions is not enough.

The system looks at both network relationships (how companies, senders, and receivers are connected) and activity over time (how their behavior changes week by week or month by month). By combining these two perspectives, the system aims to flag companies with unusual patterns that suggest potential suspicious activities, and thus deserve closer attention from the inspectors.

The following diagram shows the workflow of the anomaly detection framework:



Feature Creation: Extracting Information from the Data

The framework first transforms raw waste transport records into meaningful features that describe how each company behaves in the waste management network. These indicators are calculated weekly and monthly, and focus on two main aspects: activity

patterns (how much waste is handled and from whom), and network behavior (how companies are connected to others).

Below is a description of the features created.

Number of Individual Senders: This feature represents the count of unique individuals who sent waste to each company in a given time period. While companies can receive waste from other companies, individuals, and other types of organizations such as public institutions, individual senders are frequently involved in irregular or illegal transactions, such as the sale of stolen or improperly disposed valuable waste, or can otherwise represent informal waste management activities. A sudden increase in the number of individual senders may point to potential involvement in these illicit transactions.

Waste Quantity Received: This feature consists of the total amount of waste, in kilograms, received by each company during a given time period. Monitoring the temporal evolution of this quantity helps identify unusual spikes or drops in a company's activity that could signal suspicious behavior when combined with other features.

Waste Quantity Received from Individuals: This feature consists of the total amount of waste, in kilograms, received by each company during a given time period, specifically originating from individuals. Monitoring the temporal evolution of this quantity further helps identify suspicious patterns related to individual senders.

Betweenness Centrality: A measure of a node's importance in a network, specifically how often it lies on the shortest paths between other nodes. This feature is thus the number of shortest paths between all pairs of nodes that pass through the company at a given time period. Spikes in this feature indicate companies suddenly becoming intermediaries, which may suggest participation in illicit or waste laundering schemes.

Ego Network Density: Ego network density is a measure of how interconnected the neighbors of a focal company are within its immediate network. Formally, it is the ratio of the number of connections (transactions) among the company's neighbors to the maximum possible number of connections those nodes could share, producing a value between 0 and 1. Spikes in ego network density may indicate participation in collusive activities.

Number of Triangles: This feature consists of the sum of unique network triangles each company participates in at a given time period. Triangles are sets of three entities in a graph where each entity is directly connected to the other two, forming a closed loop. Triangles are highly suggestive of collusive activities and schemes in the network, a notion confirmed by the experience of regulatory inspectors.

Connectivity with Individuals in Triangles: This feature consists of the sum of the number of triangles that the individual waste senders connected to a company participate in, at a given time period. It measures a company's indirect exposure to suspicious behavior in the network, based on the assumption that individual senders involved in many triangles are more likely to be participating in illicit activities. This feature thus highlights companies that may be receiving stolen waste from individuals or cooperating in illicit schemes with them.

First Principal Component of Activity Features: Some detection methods can only analyze one feature at a time. To address this, several key features are combined into a single summary indicator, using Principal Component Analysis, a statistical technique that captures the joint variance of several features. This resulting feature reflects overall unusual activity by combining betweenness centrality, ego network density, number of triangles, number of individual senders, and quantity received from individuals. These features were chosen because they widely cover different aspects of the companies' network activity. Consequently, their joint variance can potentially manifest suspicious patterns in their behavior.

These features will then be used in the Statistical Detector and Deep Pattern Detector, while the Graph Embedding Detector works directly on graphs representing the waste transfer network.

Statistical Detector: Detecting Sudden Changes in Behavior

The Statistical Detector is based on the Page-Hinkley test, a statistical change detection method used to identify abrupt changes in the average behavior of a time series. For a sequence of observations, the test computes the cumulative sum of deviations from a reference value (the running mean). A change is detected when the difference between the cumulative sum and its historical minimum exceeds a predefined threshold.

This technique was applied to Monthly Connectivity with Individuals in Triangles, Weekly First Principal Component of Activity Features and Weekly Ego Density to detect anomalous deviations in these aspects of the network behavior of the companies.

The anomaly score of a company consists of the sum of the change strengths of the three highest detected changes.

Deep Pattern Detector: Detecting Unusual Temporal Patterns

The Deep Pattern detector uses a Long Short-Term Memory Variational Autoencoder (LSTM-VAE) to identify complex temporal anomalies. The LSTM component captures long-term temporal dependencies, while the Variational Autoencoder learns a compact representation of typical company activity.

This model has been previously trained to learn what normal temporal behavior looks like across weeks. It then attempts to reconstruct observed activity sequences. High reconstruction errors indicate that the observed pattern does not conform to normal behavior.

This detector is applied to weekly betweenness centrality and waste quantity received, to particularly identify cases where a company becomes highly central in the network without a corresponding increase in waste volume, which may indicate coordination or waste laundering.

The anomaly score is computed as the mean of the ten highest reconstruction errors, focusing on sporadic abnormal periods rather than overall behavior.

Graph Embedding Detector: Detecting Suspicious Network Structures

The Graph Embedding detector operates directly on the weekly waste transport network, without relying on manually engineered features.

Each weekly network is firstly processed using a Conditional Motif-aware Graph Convolutional Network (CM-GCN). A GCN is a neural network specifically designed for graph data, which learns a representation of each company by combining information from its neighboring entities. In practice, this allows the model to learn a compact summary (embedding) of each company's local network behavior, such as how connected it is and which other entities it interacts with.

In addition to standard connections, this GCN explicitly incorporates triangle motifs, which represent closed groups of three entities. These motifs are particularly relevant, as they often indicate coordination or collusion. Particularly, the model considers the connection of the companies with individual senders participating in triangles, as these are often associated with informal waste trading schemes.

The output of the GCN is a vector representation (embedding) for each company at each week. These weekly embeddings are then passed to a Long Short-Term Memory Variational Autoencoder (LSTM-VAE), similar to the one used in the Deep Pattern

Detector, to detect unusual patterns in these embeddings. The model attempts to reconstruct each company's sequence of network embeddings based on previous training. When the reconstruction error is high, it indicates that the company's network behavior deviates from learned normal patterns.

As with the Deep Pattern Detector, the anomaly score is the mean of the ten highest reconstruction errors of the company's timeseries, indicating periods of unusual behavior.

Anomaly Scoring: Combining Results From Multiple Detectors

Each detector produces its own anomaly score. Because these scores come from very different methods, they are first converted into rankings, which are then standardized to a common scale from 0 to 1, from least to most anomalous.

The final anomaly score for each company is calculated by taking the median score across the five different detectors (the Statistical Change Detector applied on the three different features, the Deep Pattern Detector and the Graph Embedding Detector). Consequently, a company exhibiting a high anomaly score in at least three of the five selected methods receives a high final anomaly score, indicating consistent suspicious behavior across multiple detection perspectives, but allowing for detection failure in at most two of the methods. Since different companies may exhibit distinct anomalous patterns, some may be flagged by certain methods but not others, highlighting the complementarity of the approaches to detect diverse forms of irregular behavior.

References

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